WEEKLY PROGRESS REPORT

Mechatronic System Design (MEC483) – Course Project

PROJECT / PHASE

Smart Pick and Place Robot

DATE RANGE

14/4/2026 – 21/4/2026

TEAM MEMBERS

Maher Abo Abed (1087993) , Sabeeha Zainab Hasham (1089042), Basel Feras Ghunaim (1088912), Ahmad Nasser Alshehhi (1090882)

HOW ARE WE DOING THIS WEEK?

TOP WINS

- Successfully built and tested and IK model using moveit and Rviz for visualizing
- Built, tested and successfully controlled a pneumatic suction system that is capable of relatively high pay loads
- Set up the Raspberry pi with the current yolov8 model, connected and set up the camera – the full system is almost in place

KEY BLOCKERS / RISKS

- Blocker: Continued printing issues – prototype and assembly can't be made yet, causing significant delays

PROJECT HEALTH

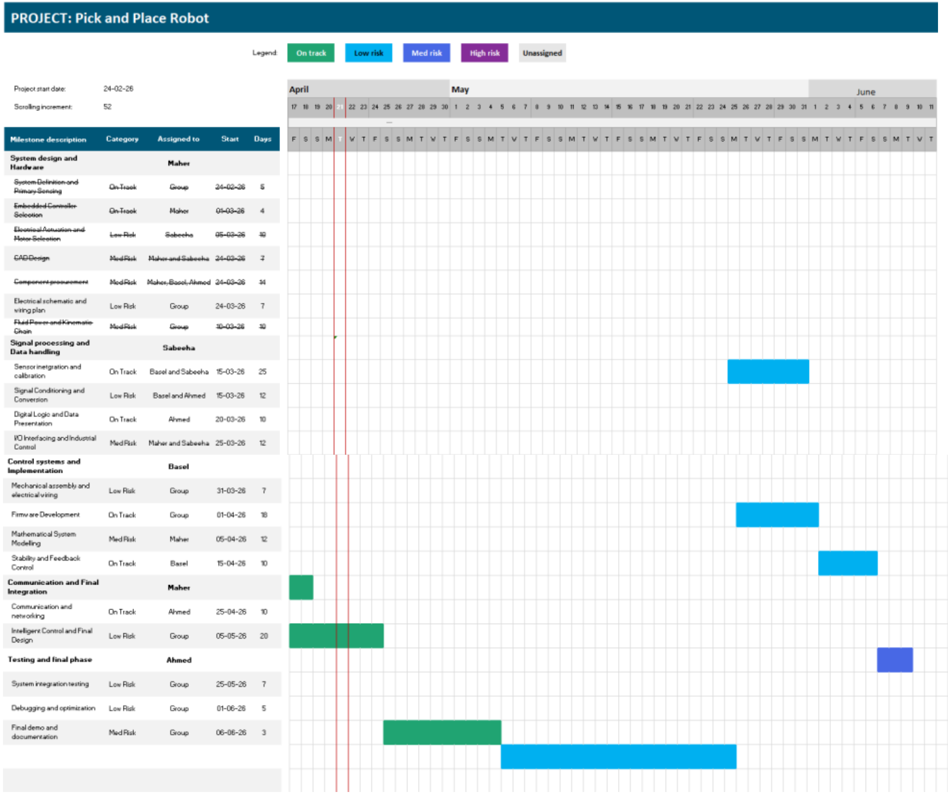
Scope	Under Execution
Schedule	Potential Delays
Fabrication & Prototyping	Under Execution

OPEN ISSUES

3 Active

Project Schedule (Gantt)

[Agile Gantt chart1.xlsx]



PROJECT: Pick and Place Robot

Legend: On track | Low risk | Med risk | High risk | Unassigned

Project start date: 24-02-26

Scrolling increment: 52

Milestone description	Category	Assigned to	Start	Days
System design and Hardware		**Maher**		
~~System Definition and Primary Sensing~~	~~On Track~~	~~Group~~	~~24-02-26~~	~~5~~
~~Embedded Controller Selection~~	~~On Track~~	~~Maher~~	~~01-03-26~~	~~4~~
~~Electrical Actuation and Motor Selection~~	~~Low Risk~~	~~Sabeeha~~	~~05-03-26~~	~~10~~
~~CAD Design~~	~~Med Risk~~	~~Maher and Sabeeha~~	~~24-03-26~~	~~7~~
~~Component procurement~~	~~Med Risk~~	~~Maher, Basel, Ahmed~~	~~24-03-26~~	~~14~~
Electrical schematic and wiring plan	Low Risk	Group	24-03-26	7
~~Fluid Power and Kinematic Chain~~	~~Med Risk~~	~~Group~~	~~10-03-26~~	~~10~~
Signal processing and Data handling		**Sabeeha**		
Sensor inetgration and calibration	On Track	Basel and Sabeeha	15-03-26	25
Signal Conditioning and Conversion	Low Risk	Basel and Ahmed	15-03-26	12
Digital Logic and Data Presentation	On Track	Ahmed	20-03-26	10
I/O Interfacing and Industrial Control	Med Risk	Maher and Sabeeha	25-03-26	12
Control systems and Implementation		**Basel**		
Mechanical assembly and electrical wiring	Low Risk	Group	31-03-26	7
Firmware Development	On Track	Group	01-04-26	18
Mathematical System Modelling	Med Risk	Maher	05-04-26	12
Stability and Feedback Control	On Track	Basel	15-04-26	10
Communication and Final Integration		**Maher**		
Communication and networking	On Track	Ahmed	25-04-26	10
Intelligent Control and Final Design	Low Risk	Group	05-05-26	20
Testing and final phase		**Ahmed**		
System integration testing	Low Risk	Group	25-05-26	7
Debugging and optimization	Low Risk	Group	01-06-26	5
Final demo and documentation	Med Risk	Group	06-06-26	3

ISSUES, RISKS, AND ACTION ITEMS

TOP ISSUES (BLOCKING PROGRESS)

- Fabrication delays
- Significant halt

MITIGATIONS / DECISIONS NEEDED

- Constant printing supervision is needed – problem is printers are not located here

ACTION ITEMS (NEXT 7 DAYS)

ID	ACTION ITEM DESCRIPTION	MEMBER	WEEK DUE
A1	TBD	Sabeeha	Week 8
A2	TBD	Maher	Week 8
A3	TBD	Basel	Week 8
A4	TBD	Ahmad	Week 8

INDIVIDUAL UPDATES

WEEKLY PROGRESS & NEXT STEPS

MEMBER 1
Maher Abo Abed

MEMBER 2
Sabeeha Zainab Hasham

MEMBER 3
Basel Feras Ghunaim

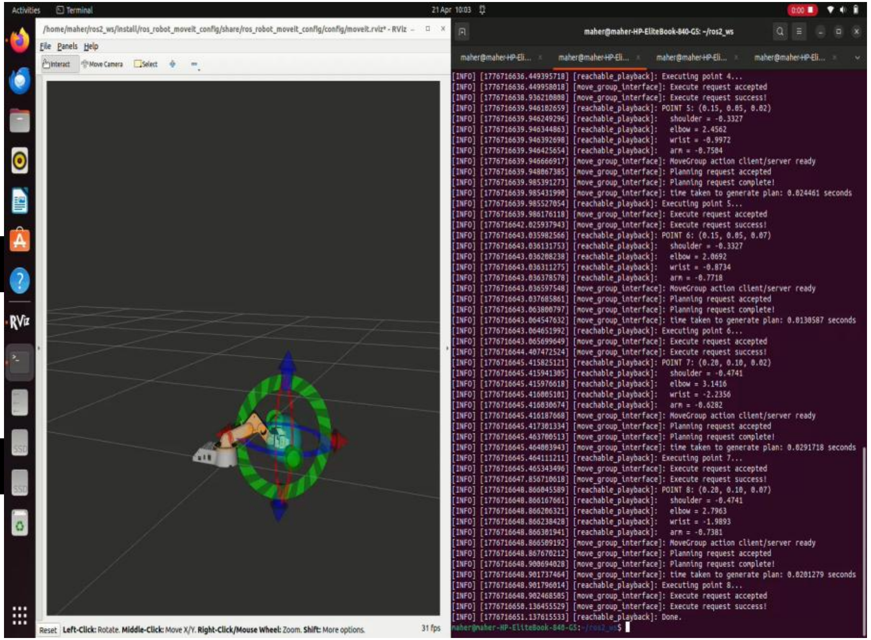
MEMBER 4
Ahmad Nasser Alshehhi

MEMBER 1: Maher Abo Abed

ACCOMPLISHED (SINCE LAST REPORT)

- Finished IK (ROS Based)
  - IKFast – Not successful (older solution, requires almost perfect urdf, visualizing is difficult)
  - Hand calculations – attempted but preferred other solutions
  - Moveit – working, trial and tested, can be visualized
- Analyzed the reachable workspace using this newly made model
- Attempted random poses, and verified solution
- Started with object tracking/following
- Made a successful code, but suffered with hardware issues

EVIDENCE

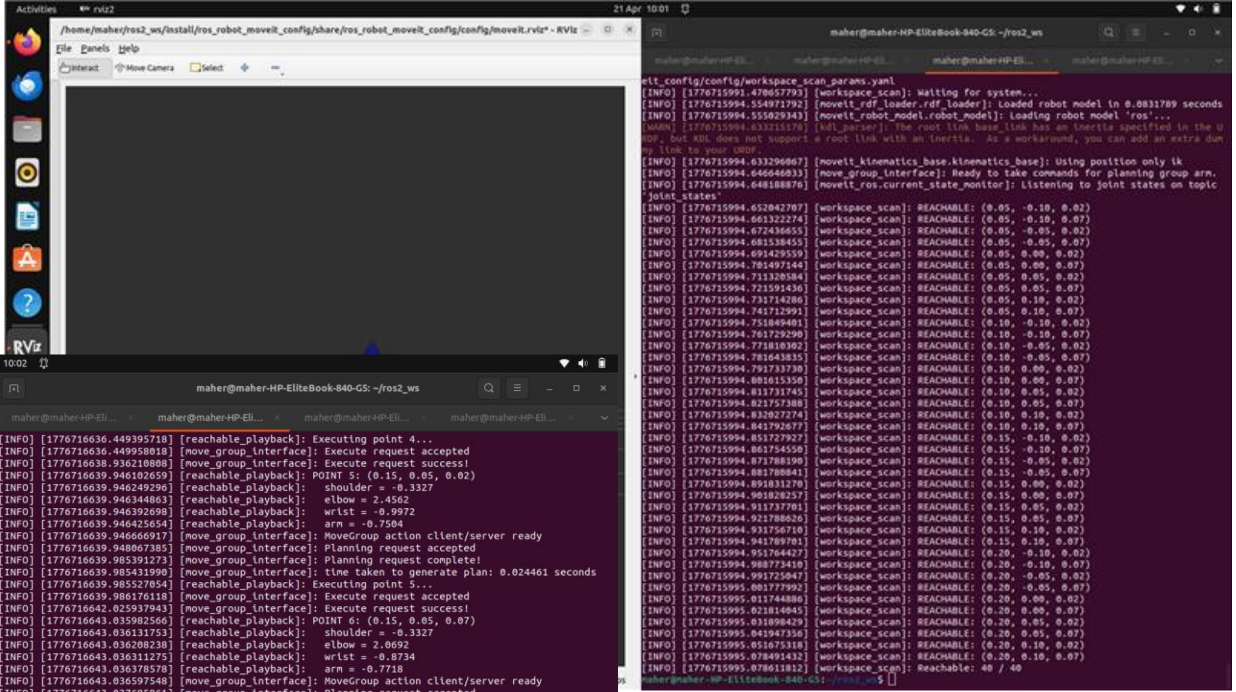


IN PROGRESS

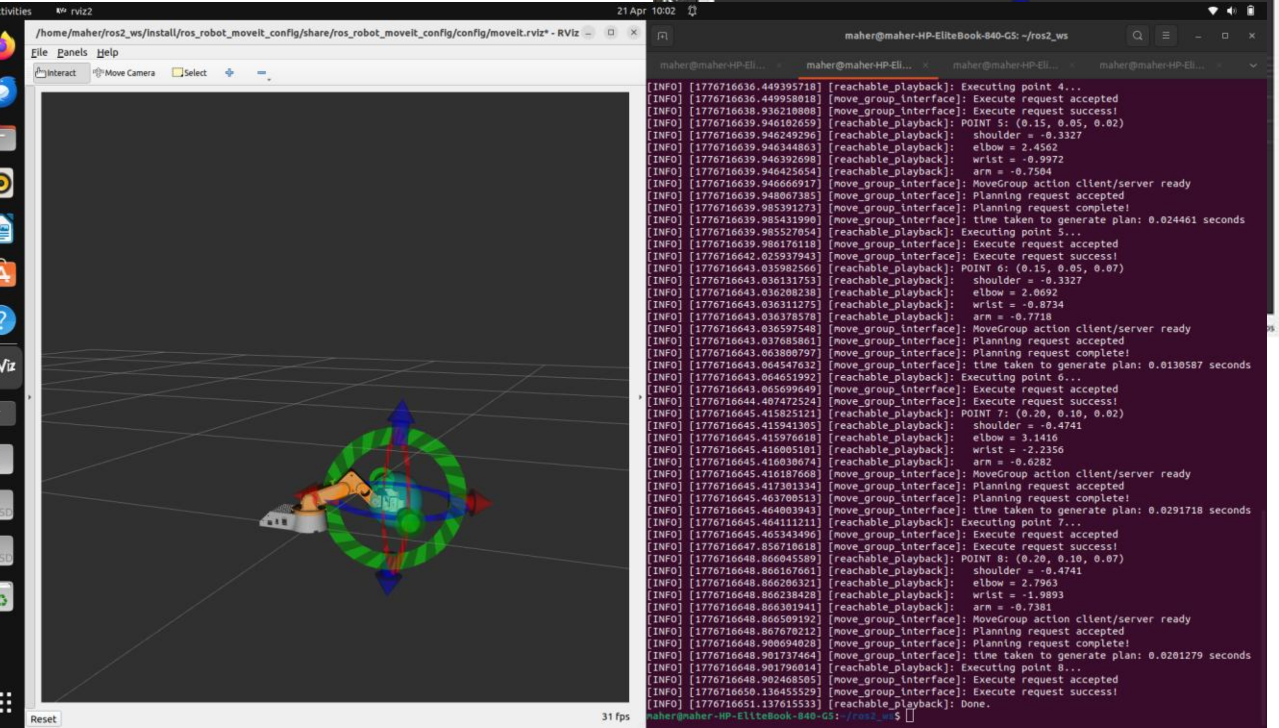
- Inverse kinematic for the suction
- Object tracking and following
- Awaiting printed parts

NEXT WEEK PLAN

- Mechanical assembly
- Finish the IK
- Finish the object tracking/following.



Activities
rviz2
21 Apr 10:01
/home/maher/ros2_ws/install/ros_robot_moveit_config/share/ros_robot_moveit_config/config/moveit.rviz* - RViz
File Panels Help
Interact Move Camera Select
maher@maher-HP-EliteBook-840-G5: ~/ros2_ws
elt_config/config/workspace_scan_params.yaml
[INFO] [1776715991.470657793] [workspace_scan]: Waiting for system...
[INFO] [1776715994.554971792] [moveit_rdf_loader.rdf_loader]: Loaded robot model in 0.0831789 seconds
[INFO] [1776715994.555029343] [moveit_robot_model.robot_model]: Loading robot model 'ros'...
[WARN] [1776715994.633215170] [kdl_parser]: The root link base_link has an inertia specified in the URDF, but KDL does not support a root link with an inertia. As a workaround, you can add an extra dummy link to your URDF.
[INFO] [1776715994.633296067] [moveit_kinematics_base.kinematics_base]: Using position only ik
[INFO] [1776715994.646646033] [move_group_interface]: Ready to take commands for planning group arm.
[INFO] [1776715994.648188876] [moveit_ros.current_state_monitor]: Listening to joint states on topic 'joint_states'
[INFO] [1776715994.652042707] [workspace_scan]: REACHABLE: (0.05, -0.10, 0.02)
[INFO] [1776715994.661322274] [workspace_scan]: REACHABLE: (0.05, -0.10, 0.07)
[INFO] [1776715994.672436655] [workspace_scan]: REACHABLE: (0.05, -0.05, 0.02)
[INFO] [1776715994.681538455] [workspace_scan]: REACHABLE: (0.05, -0.05, 0.07)
[INFO] [1776715994.691429559] [workspace_scan]: REACHABLE: (0.05, 0.00, 0.02)
[INFO] [1776715994.701497144] [workspace_scan]: REACHABLE: (0.05, 0.00, 0.07)
[INFO] [1776715994.711320584] [workspace_scan]: REACHABLE: (0.05, 0.05, 0.02)
[INFO] [1776715994.721591436] [workspace_scan]: REACHABLE: (0.05, 0.05, 0.07)
[INFO] [1776715994.731714286] [workspace_scan]: REACHABLE: (0.05, 0.10, 0.02)
[INFO] [1776715994.741712991] [workspace_scan]: REACHABLE: (0.05, 0.10, 0.07)
[INFO] [1776715994.751849401] [workspace_scan]: REACHABLE: (0.10, -0.10, 0.02)
[INFO] [1776715994.761729290] [workspace_scan]: REACHABLE: (0.10, -0.10, 0.07)
[INFO] [1776715994.771810302] [workspace_scan]: REACHABLE: (0.10, -0.05, 0.02)
[INFO] [1776715994.781643835] [workspace_scan]: REACHABLE: (0.10, -0.05, 0.07)
[INFO] [1776715994.791733730] [workspace_scan]: REACHABLE: (0.10, 0.00, 0.02)
[INFO] [1776715994.801615350] [workspace_scan]: REACHABLE: (0.10, 0.00, 0.07)
[INFO] [1776715994.811731745] [workspace_scan]: REACHABLE: (0.10, 0.05, 0.02)
[INFO] [1776715994.821757388] [workspace_scan]: REACHABLE: (0.10, 0.05, 0.07)
[INFO] [1776715994.832027274] [workspace_scan]: REACHABLE: (0.10, 0.10, 0.02)
[INFO] [1776715994.841792677] [workspace_scan]: REACHABLE: (0.10, 0.10, 0.07)
[INFO] [1776715994.851727927] [workspace_scan]: REACHABLE: (0.15, -0.10, 0.02)
[INFO] [1776715994.861754550] [workspace_scan]: REACHABLE: (0.15, -0.10, 0.07)
[INFO] [1776715994.871788190] [workspace_scan]: REACHABLE: (0.15, -0.05, 0.02)
[INFO] [1776715994.881788841] [workspace_scan]: REACHABLE: (0.15, -0.05, 0.07)
[INFO] [1776715994.891831270] [workspace_scan]: REACHABLE: (0.15, 0.00, 0.02)
[INFO] [1776715994.901828257] [workspace_scan]: REACHABLE: (0.15, 0.00, 0.07)
[INFO] [1776715994.911737701] [workspace_scan]: REACHABLE: (0.15, 0.05, 0.02)
[INFO] [1776715994.921788626] [workspace_scan]: REACHABLE: (0.15, 0.05, 0.07)
[INFO] [1776715994.931756710] [workspace_scan]: REACHABLE: (0.15, 0.10, 0.02)
[INFO] [1776715994.941789701] [workspace_scan]: REACHABLE: (0.15, 0.10, 0.07)
[INFO] [1776715994.951764427] [workspace_scan]: REACHABLE: (0.20, -0.10, 0.02)
[INFO] [1776715994.988773410] [workspace_scan]: REACHABLE: (0.20, -0.10, 0.07)
[INFO] [1776715994.991725047] [workspace_scan]: REACHABLE: (0.20, -0.05, 0.02)
[INFO] [1776715995.001777992] [workspace_scan]: REACHABLE: (0.20, -0.05, 0.07)
[INFO] [1776715995.011744886] [workspace_scan]: REACHABLE: (0.20, 0.00, 0.02)
[INFO] [1776715995.021814045] [workspace_scan]: REACHABLE: (0.20, 0.00, 0.07)
[INFO] [1776715995.031898429] [workspace_scan]: REACHABLE: (0.20, 0.05, 0.02)
[INFO] [1776715995.041947356] [workspace_scan]: REACHABLE: (0.20, 0.05, 0.07)
[INFO] [1776715995.051675318] [workspace_scan]: REACHABLE: (0.20, 0.10, 0.02)
[INFO] [1776715995.078491432] [workspace_scan]: REACHABLE: (0.20, 0.10, 0.07)
[INFO] [1776715995.078611812] [workspace_scan]: Reachable: 40 / 40



Activities
rviz2
21 Apr 10:02
/home/maher/ros2_ws/install/ros_robot_moveit_config/share/ros_robot_moveit_config/config/moveit.rviz* - RViz
File Panels Help
Interact Move Camera Select
Reset
31 fps
maher@maher-HP-EliteBook-840-G5: ~/ros2_ws
[INFO] [1776716636.449395718] [reachable_playback]: Executing point 4...
[INFO] [1776716636.449958018] [move_group_interface]: Execute request accepted
[INFO] [1776716638.936210808] [move_group_interface]: Execute request success!
[INFO] [1776716639.946102659] [reachable_playback]: POINT 5: (0.15, 0.05, 0.02)
[INFO] [1776716639.946249296] [reachable_playback]: shoulder = -0.3327
[INFO] [1776716639.946344863] [reachable_playback]: elbow = 2.4562
[INFO] [1776716639.946392698] [reachable_playback]: wrist = -0.9972
[INFO] [1776716639.946425654] [reachable_playback]: arm = -0.7504
[INFO] [1776716639.946666917] [move_group_interface]: MoveGroup action client/server ready
[INFO] [1776716639.948067385] [move_group_interface]: Planning request accepted
[INFO] [1776716639.985391273] [move_group_interface]: Planning request complete!
[INFO] [1776716639.985431990] [move_group_interface]: time taken to generate plan: 0.024461 seconds
[INFO] [1776716639.985527054] [reachable_playback]: Executing point 5...
[INFO] [1776716639.986176118] [move_group_interface]: Execute request accepted
[INFO] [1776716642.025937943] [move_group_interface]: Execute request success!
[INFO] [1776716643.035982566] [reachable_playback]: POINT 6: (0.15, 0.05, 0.07)
[INFO] [1776716643.036131753] [reachable_playback]: shoulder = -0.3327
[INFO] [1776716643.036208238] [reachable_playback]: elbow = 2.0692
[INFO] [1776716643.036311275] [reachable_playback]: wrist = -0.8734
[INFO] [1776716643.036378578] [reachable_playback]: arm = -0.7718
[INFO] [1776716643.036597548] [move_group_interface]: MoveGroup action client/server ready
[INFO] [1776716643.037685861] [move_group_interface]: Planning request accepted
[INFO] [1776716643.063800797] [move_group_interface]: Planning request complete!
[INFO] [1776716643.064547632] [move_group_interface]: time taken to generate plan: 0.0130587 seconds
[INFO] [1776716643.064651992] [reachable_playback]: Executing point 6...
[INFO] [1776716643.065699649] [move_group_interface]: Execute request accepted
[INFO] [1776716644.407472524] [move_group_interface]: Execute request success!
[INFO] [1776716645.415825121] [reachable_playback]: POINT 7: (0.20, 0.10, 0.02)
[INFO] [1776716645.415941305] [reachable_playback]: shoulder = -0.4741
[INFO] [1776716645.415976618] [reachable_playback]: elbow = 3.1416
[INFO] [1776716645.416005101] [reachable_playback]: wrist = -2.2356
[INFO] [1776716645.416030674] [reachable_playback]: arm = -0.6282
[INFO] [1776716645.416187668] [move_group_interface]: MoveGroup action client/server ready
[INFO] [1776716645.417301334] [move_group_interface]: Planning request accepted
[INFO] [1776716645.463700513] [move_group_interface]: Planning request complete!
[INFO] [1776716645.464003943] [move_group_interface]: time taken to generate plan: 0.0291718 seconds
[INFO] [1776716645.464111211] [reachable_playback]: Executing point 7...
[INFO] [1776716645.465343496] [move_group_interface]: Execute request accepted
[INFO] [1776716647.856710618] [move_group_interface]: Execute request success!
[INFO] [1776716648.860045589] [reachable_playback]: POINT 8: (0.20, 0.10, 0.07)
[INFO] [1776716648.866167661] [reachable_playback]: shoulder = -0.4741
[INFO] [1776716648.866206321] [reachable_playback]: elbow = 2.7963
[INFO] [1776716648.866238428] [reachable_playback]: wrist = -1.9893
[INFO] [1776716648.866301941] [reachable_playback]: arm = -0.7381
[INFO] [1776716648.866509192] [move_group_interface]: MoveGroup action client/server ready
[INFO] [1776716648.867670212] [move_group_interface]: Planning request accepted
[INFO] [1776716648.900694028] [move_group_interface]: Planning request complete!
[INFO] [1776716648.901737464] [move_group_interface]: time taken to generate plan: 0.0201279 seconds
[INFO] [1776716648.901796014] [reachable_playback]: Executing point 8...
[INFO] [1776716648.902468505] [move_group_interface]: Execute request accepted
[INFO] [1776716650.136455529] [move_group_interface]: Execute request success!
[INFO] [1776716651.137615533] [reachable_playback]: Done.
maher@maher-HP-EliteBook-840-G5:~/ros2_ws$

```python
def main():
    print(f"Target class: {TARGET_CLASS_NAMES}")

    model = YOLO(MODEL_PATH)

    ser = serial.Serial(SERIAL_PORT, BAUD_RATE, timeout=1)
    time.sleep(2)

    cap = cv2.VideoCapture(CAMERA_INDEX)

    last_cmd = "S"

    while True:
        ret, frame = cap.read()
        if not ret:
            break

        frame_height, frame_width = frame.shape[:2]

        results = model(frame, conf=CONFIDENCE_THRESHOLD, verbose=False)
        result = results[0]

        annotated_frame = result.plot()

        target = choose_target(result, model)

        cmd, error_x = decide_command(frame_width, target)

        last_cmd = send_command(ser, cmd, last_cmd)

        if target:
            cv2.putText(annotated_frame, f"Target: apple", (10, 30),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,255,0), 2)
        else:
            cv2.putText(annotated_frame, "Target: NONE", (10, 30),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0,0,255), 2)

        cv2.putText(annotated_frame, f"Command: {cmd}", (10, 70),
                    cv2.FONT_HERSHEY_SIMPLEX, 0.9, (255,255,0), 2)

        cv2.imshow("Apple Follower", annotated_frame)

        if cv2.waitKey(1) & 0xFF == ord("q"):
            break

    ser.write(b"S")
    ser.close()
```

Restricted Mode ⓧ 0 ⚠ 3 Ln 108, Col 1 Spaces: 4 UTF-8 LF { } Python

- The program uses a camera to detect objects in real time using a YOLOv8 model.
- It is specifically configured to detect only apples.For every video frame, the model finds objects and filters out everything except apples with sufficient confidence.If multiple apples are detected, the program selects the largest one (closest to the camera)
- It calculates the horizontal position (center) and size (area) of the selected apple.
- The program compares the apple's position to the center of the screen.
- If the apple is to the left, it sends a "L" command (turn left).If the apple is to the right, it sends a "R" command (turn right).
- If the apple is centered:It sends "F" (move forward) if the apple is far away.It sends "S" (stop) if the apple is close enough.
- Commands are sent through a serial connection to a device like an Arduino.The program avoids sending the same command repeatedly unless it changes.
- It displays the video feed with detection boxes and shows the current target and command on screen.
- The loop runs continuously until the user presses "q" to quit.
- When exiting, it sends a stop command, closes the serial connection, and releases the camera

MEMBER 2: Sabeeha Hasham

ACCOMPLISHED (SINCE LAST REPORT)

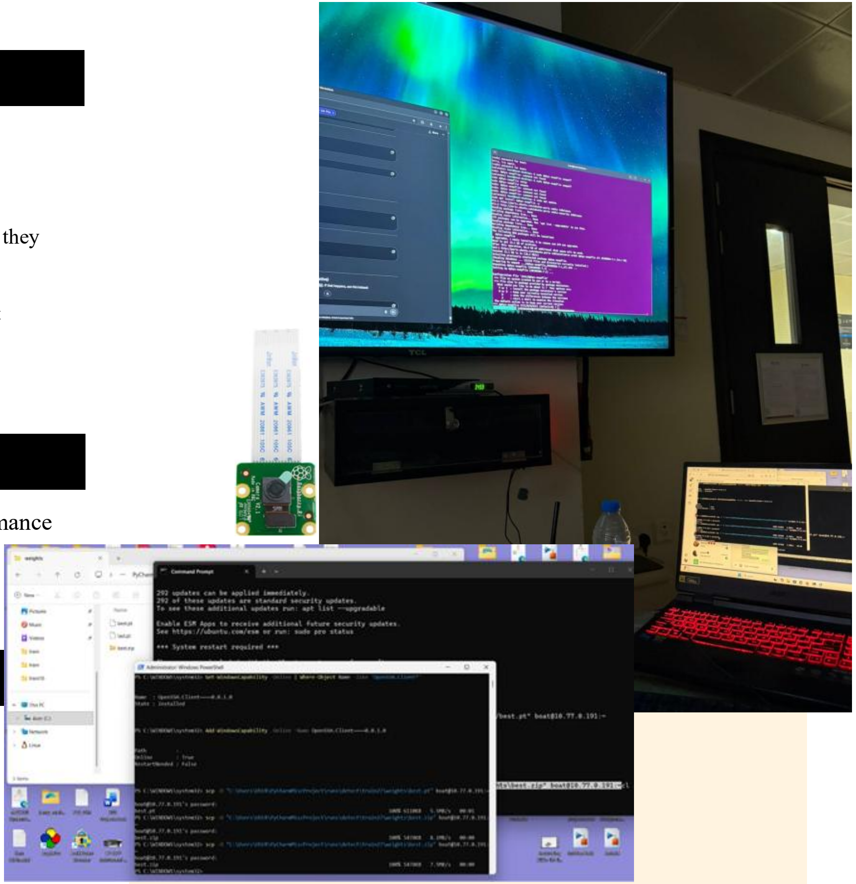
- Installed or worked on camera-related tools (like v4l-utils) for camera support
  - Connected the Raspberry Pi Camera Module 2 to the Pi
- Used Windows PowerShell with the scp command to directly transfer best.zip from my PC to the Raspberry Pi over the network, instead of using file-sharing apps like WhatsApp or Google Drive – they didn't work
- Attempted to run your object detection system using YOLOv8
  - Encountered an error: "Illegal instruction (core dumped)" the model or environment isn't compatible with the Pi setup
- Worked inside a Python virtual environment (yolovenv
- Adjusted system settings (like memory/swap limits) to improve performance

IN PROGRESS

- Camera is physically connected but the detection code crashes due to compatibility/performance issues

NEXT WEEK PLAN

EVIDENCE (SHOW, DON'T TELL)



MEMBER 3: Basel Feras Ghunaim

ACCOMPLISHED (SINCE LAST REPORT)

- Completed component selection and purchasing process for the suction system Acquired:
  - Vacuum pump
  - Suction cup
  - Tubing and connectors
- Successfully assembled and tested the suction system
- Achieved lifting capability of up to 200 g, exceeding initial requirements
- Validated that the suction system is functional and reliable for target objects

NEXT WEEK PLAN

- Explore and implement remote control of the robotic arm/gripper.
- Investigate possible control methods:
  - Wireless control (Bluetooth / Wi-Fi)
  - Remote interface via Raspberry Pi or Arduino
- Select the most suitable approach and begin initial implementation and testing

IN PROGRESS

- Evaluating overall system integration (gripper + suction)
- Observing performance and identifying any improvements in stability or sealing
- Discussing control methods for remote operation as a team

MEMBER 4: Ahmed Alshehhi

ACCOMPLISHED (SINCE LAST REPORT)

- Looked into basic Arduino connections for joystick control
- Reviewed simple circuit examples to understand wiring layout.

IN PROGRESS

- Preparing the wiring setup for the joystick and Arduino
- Understanding signal mapping between joystick input and motor movement

NEXT WEEK PLAN

- Begin assembling the control components.
- Start initial testing of joystick input with Arduino.

HELP NEEDED / QUESTIONS

-

WRAP-UP: WHAT WE NEED FROM THE INSTRUCTOR / STAKEHOLDERS

DECISIONS REQUESTED

QUESTIONS / FEEDBACK

ACTION ITEMS FOLLOW-UP

APPENDIX

- https://maherrrrr99.github.io/maherrrrr99/
- https://sabeehahasham.github.io/
- https://basel-ghunaim.github.io/
- https://ahmed1090822.github.io/Ahmed109.github.io/